

Predicting White Elephant Risk in Olympic and World Cup Venues Using Machine Learning

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Date: August 28, 2026

Synopsis

Cities that host the Olympics or the World Cup build venues that see a few weeks of heavy use and then have to justify their upkeep for decades. We asked whether the information a city already has at the planning stage, before anything is built, can flag which venues will end up abandoned. Working from 875 Olympic venues, 26 of which were eventually abandoned, we found that it cannot, at least not from the planning-stage information we were able to assemble. The most useful result we have is the reason why: the feature our reasoning leaned on hardest, whether a venue was built for a single sport or designed for mixed use afterwards, is not recorded for Olympic venues, and the nearest stand-in the data offers describes what a venue was used for rather than what it was built for.

Introduction

Background

Cities that host the Olympics or the World Cup commit public money to venues years before those venues open. Some go on to serve their host cities for decades; others sit empty within a few years, consuming maintenance costs with little public benefit, “white elephants.” Hosting already runs over budget more than any other type of megaproject, by an average of 156% in real terms, and every Games since 1960 has cost more than planned (Flyvbjerg et al., 2016; Flyvbjerg et al., 2024). Venues absorb a large share of that spending, and FIFA and the IOC take the event’s revenue while host cities carry the venue costs and the aftermath (Alm et al., 2016).

Existing research on this problem is almost entirely retrospective. Müller (2015) documents a recurring pattern across Olympic host cities he calls the “mega-event syndrome”: compressed timelines, inflated cost estimates, and limited oversight that push host cities toward oversized, single-purpose venues with little post-event use. Alm et al. (2016) measured how venues from 1996 to 2010 were used after their event and found ownership, capacity, and corruption mattered, but every venue in that study was newly built for its event, so it could not compare new venues against existing ones. No study we are aware of trains a predictive model on pre-construction information and tests it against venues it has never seen. Host-city bid committees and government planning offices currently have no data-driven early-warning tool for flagging a proposed venue as high-risk before groundbreaking, only after-the-fact case studies of individual failures. This project addresses that gap, for a stakeholder, host-city bid committees and government planning offices, that is politically accountable and mostly non-technical and needs a model whose reasoning it can read and defend in public.

Host-country income is one of our predictors, and it raises an ethical concern worth naming directly. Read carelessly, a model like ours could be taken to argue that poorer countries should not host. That is not the intent. The model is a planning tool meant to flag a risky venue plan early enough to change it, by building smaller, planning a second use, or using a stadium that already exists, rather than a reason to reject a bid. The same caution applied to host-country corruption, which we intended to use and dropped for lack of coverage.

Question

Can information a host city already has at the planning stage, before anything is built, predict whether a venue stays in active use or becomes a white elephant?

The question is unchanged since the proposal, but what we mean by planning-stage information has narrowed. We set out to use design purpose, capacity, event type, host-country income and corruption, newly built status, and distance from the host city center. Three of those reached the final models: capacity, host-country income, and newly built status. Corruption was dropped because the index we could obtain only begins in 2012, which is after most of the venues in our sample were built. Distance from the city center could not be established for more than three quarters of Olympic venues, and design purpose is not recorded for them at all. Event type did not enter as a predictor either, because the two events were modeled separately rather than pooled. In place of what fell out we added two predictors we had not originally planned on, host-city population and the region the host city sits in.

Hypothesis

Venues that are newly built, single-purpose, and far from the host city center become white elephants more often, because long-term use was never the point of building them.

The hypothesis is also unchanged, and we were only able to test one of its three parts. Newly built status is in the models. Single-purpose design and distance from the city center are the two features we could not measure for Olympic venues, so nothing in our results should be read as evidence for or against them.

Prediction

Newly built status and design purpose will emerge as the strongest predictors, and the model will outperform a naive baseline that always predicts a venue stays active.

The prediction stands as we made it, and both halves of it turned out wrong. Design purpose could not be measured for Olympic venues at all, new-build status scored lowest of the five predictors in the random forest, and no model beat a naive baseline that calls every venue safe. Discussion and Next Steps takes up what that tells us.

Data

Data Acquisition

The dataset combines Olympic and World Cup venue records with country and city-level reference data. We chose this combination because it captures information available before a venue is built, which is exactly the window our research question asks about. Predictor data comes from four external sources. The World Bank, via the WDI package, supplies GDP per capita from 1960 to 2026. Transparency International supplies the Corruption Perceptions Index. That index only goes back to 2012, so it doesn't exist before then; we dropped it as a predictor in Sprint 4 for that reason. The UN World Population Prospects supplies city population figures from 1975 to 2050, capped at 2026 for this project. Wikidata supplies venue coordinates, seating capacity, and opening or inception year, queried via SPARQL. We matched Olympic venues to Wikidata through the Wikipedia Search API, since IOC records carry no direct links to Wikidata entries.

The target variable flags whether a venue became a white elephant. It comes from records the team compiled and hand-coded, documenting each venue's current status and, where applicable, the reason it fell out of use. Of 983 raw Olympic rows, 877 carry no reason code, since a reason is only recorded for venues that closed. Of those 877, 769 are venues still in use, which need no

reason and are valid non-failure rows in the analytical sample. The remaining 108 are temporary structures built for the Games, which are dropped entirely rather than modeled. The other 106 raw rows do carry an explicit reason: 69 lost to urban redevelopment, 6 replaced by a successor stadium, and 5 to war, none of which count as the kind of failure this project studies, plus the 26 rows coded as abandoned before demolition, our true positives. Together, the 769 in-use rows and the 106 reason-coded rows make up the analytical Olympic sample: 875 venues with 26 failures, about 3.0%.

Two known issues in the source data are worth flagging up front; we cover them more fully in Discussion and Next Steps. First, the Tokyo Olympic Village includes two distinct buildings that share a name and host city. The current train/test split treats them as one venue. Second, the target reflects venue status only as of data collection. A venue currently coded as in active use may still fail later. That's a form of right-censoring the model can't account for.

Cleaning

Data cleaning was built around one central concern: preventing information from outside a venue's own pre-construction profile from leaking into how the model is trained or tested. GDP and city population are ranked as percentiles against their full external reference panels, the entire World Bank GDP panel and the entire UN city population panel, rather than only the small set of project host cities and countries, and this ranking happens before the train/test split, because ranking against an external, population-level panel cannot leak test-set information. Capacity has no comparable external panel to rank against, so its percentile rank is computed only after the split, using a custom function (`rank_within_train()`) that ranks each value only against the training distribution, so the training data never sees test values. Imputation of missing values likewise uses train-only medians for the same reason. The train/test split itself is stratified

and grouped by venue, with a data-derived ratio of 1 over the square root of the number of predictors.

Several real bugs were caught and fixed during cleaning, documented here rather than silently corrected, since each one changed a reported number. A GDP percentile-rank degeneracy, caused by ranking within a single award year against too small a comparison group, collapsed most ranks to near zero; it was fixed by ranking against the full World Bank panel instead. The same bug recurred independently in the city population percentile rank and was fixed the same way. A Youden-threshold direction bug, caught through the reasoning that Youden's J cannot fall below zero at a true optimum, traced back to the ROC calculation letting direction be auto-detected from weak, noisy data rather than fixed explicitly; adding an explicit direction argument corrected it. And one purely textual inconsistency, a stated sample size of 983 that should have read 875 after filtering, was corrected in the report text with no rerun required.

One predictor, venue age, was tested and deliberately dropped rather than kept by default. A Fisher's exact test comparing each imputed predictor's missing-value flag against the outcome found that every one of the 17 training failures had a missing venue age, versus 0 of 110 venues with a known venue age, a near-complete confound indicating the variable was effectively measuring whether a Wikidata record existed for that venue rather than the venue's actual age. It was dropped entirely and replaced with host-city population percentile rank; re-running confirmed the model's AUC barely moved, itself a useful result, since it showed venue age was never contributing real signal to begin with.

Data Exploration

The exploratory work in our Sprint 3 report describes a different outcome from the one we ended up modeling, so it is not carried forward. Sprint 3 derived its target inline, counting any venue

not currently in use as a failure. But the target was frozen later, once the reason coding was complete, to count only venues abandoned before demolition. Redevelopment, replacement by a successor stadium, and war are recorded as reasons a venue left use but are not the kind of failure this project studies. On the same 875 venues the two rules disagree about 80 of them, 12.1% against 3.0% (Figure 1). The Sprint 3 figures are not stale numbers about our outcome but accurate numbers about a different one. Everything in this section was rebuilt on the dataset the models are actually fitted on, after cleaning, feature engineering, the train/test split, and imputation.

Table 1 reconciles the raw pull with the sample we model. The Olympic pull holds 983 venue records. Temporary venues, 108 of them, have no coded outcome, because a structure dismantled on schedule is neither a success nor a failure in the sense we are predicting, and they drop out at the split. That leaves an analytical sample of 875 venues carrying 26 failures, a rate of 3.0%. The split holds that rate steady, 567 training venues at 3.0% and 308 test venues at 2.9%.

We explore all 875 rows here rather than the 567 training rows alone. With 26 failures in total, restricting the outcome-stratified figures to the training set would leave 17 spread across four regions and three eras, and several cells would fall to one or two venues. The models were still fitted on training rows only, and the imputation medians still come from training rows only.

The most important thing this dataset does is go missing. Table 2 gives the completeness of each predictor as it stood before imputation. Seating capacity is absent for 84.5% of the sample, venue age and years since construction for 80.8%, and distance from the city center for 77.4%. GDP is absent for 38.2% and host-city population for 13.5%. Only region and years since the event are complete. The missingness is not random: it tracks whether a venue has a Wikidata record at all,

which is why testing venue age against the outcome found that every one of the 17 training failures had a missing venue age, and why that predictor was dropped.

Figure 2 shows what imputation does to the column it affects most. Before imputation, the 136 venues with a recorded capacity spread from about 100 seats to about 93,000. After imputation, 739 of 875 venues carry the same value, the training median of 9.63 on the log scale, about 15,200 seats. The variable the models see is therefore a real measurement for fewer than one venue in six and a constant for the rest. Its standard deviation falls from 1.14 among the observed values to 0.45 across the imputed column as a whole. Discussion and Next Steps returns to this limitation, which is the strongest single reason to treat capacity's apparent importance in any of our models with suspicion.

Figures 3 to 5 stratify the outcome by the three categorical predictors we can read directly. Venues built for the Games fail at 3.8%, against 2.3% for venues that already existed (Figure 3). The direction matches our hypothesis, but the gap rests on 15 failures against 11, and no model in this project recovered it as a usable signal. By region, Europe runs highest at 4.0% and Asia lowest at 1.7%, with Oceania at zero across 47 venues (Figure 4). At an overall rate of 3% a group that size would be expected to contain one or two failures, so the zero carries no weight on its own. By era, venues from the 1896 to 1936 Games fail at 5.0% against 2.0% for 1948 to 2000 and 3.9% for 2002 to 2022 (Figure 5). The oldest group has had the most time in which a venue could fall out of use, so these three rates are not directly comparable, and we do not read a trend into them.

Taken together the exploration says the same thing the modeling later said. The class is small and unbalanced at 26 failures, the predictor with the most theoretical weight behind it is largely

absent from the data, and the differences visible between groups are the size that 26 events spread across four regions and three eras would produce by chance.

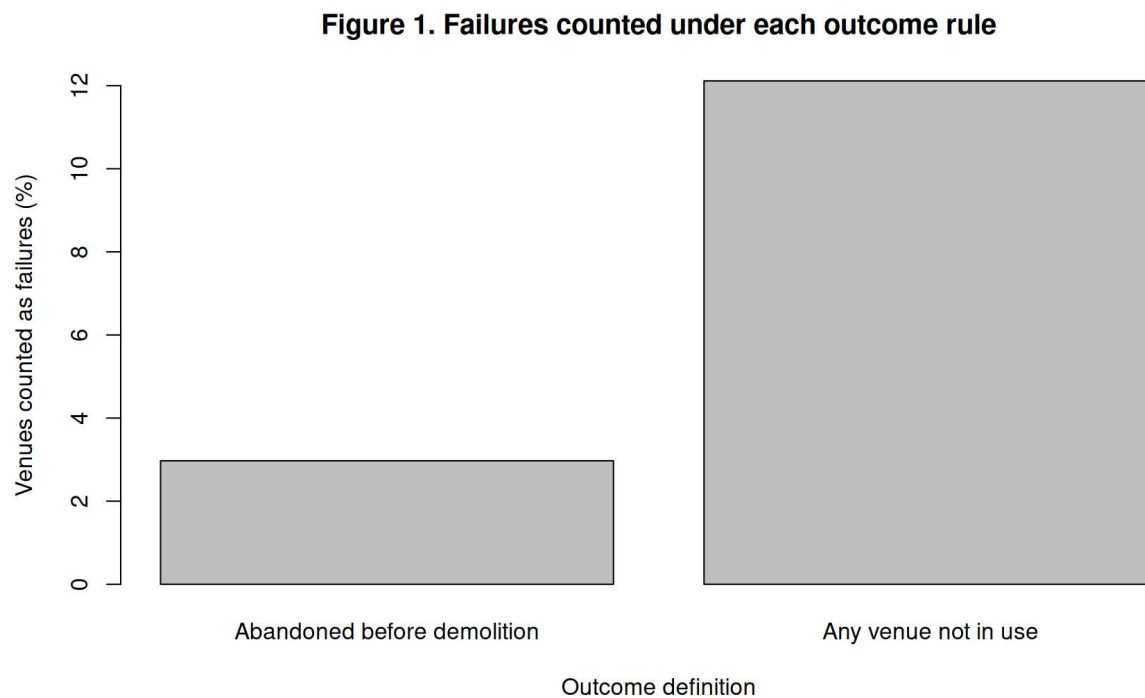


Figure 1. Venues counted as failures under the frozen outcome definition and under the definition used in the Sprint 3 report, on the same 875 Olympic venues.

Figure 2a. Capacity before imputation, 136 observed venues

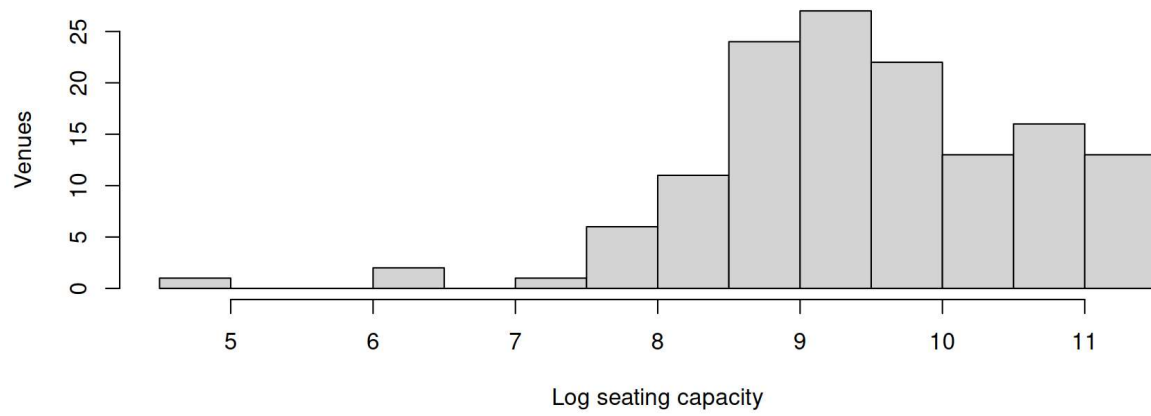


Figure 2b. Capacity after imputation, all 875 venues

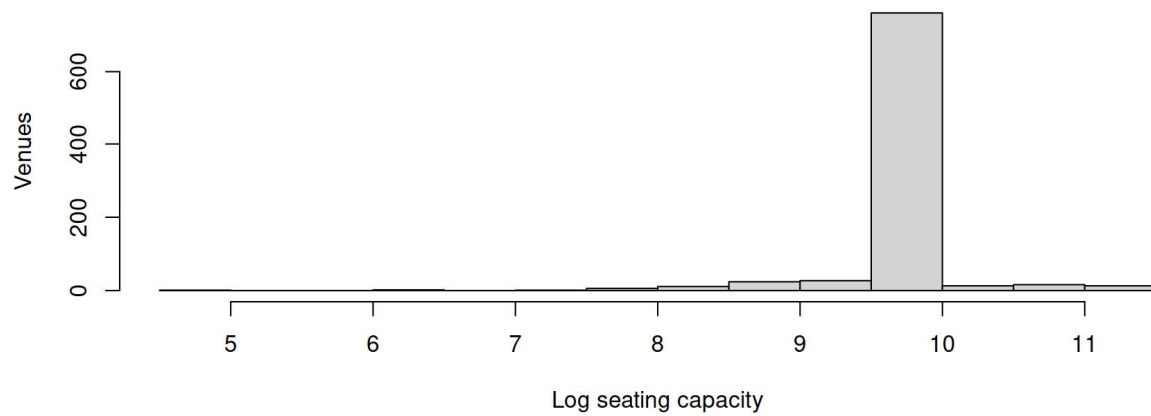


Figure 2. Distribution of log seating capacity before imputation (panel 2a, the 136 venues with a recorded capacity) and after imputation (panel 2b, all 875). The two panels use different vertical scales.

Figure 3. Failure rate by whether the venue was built new

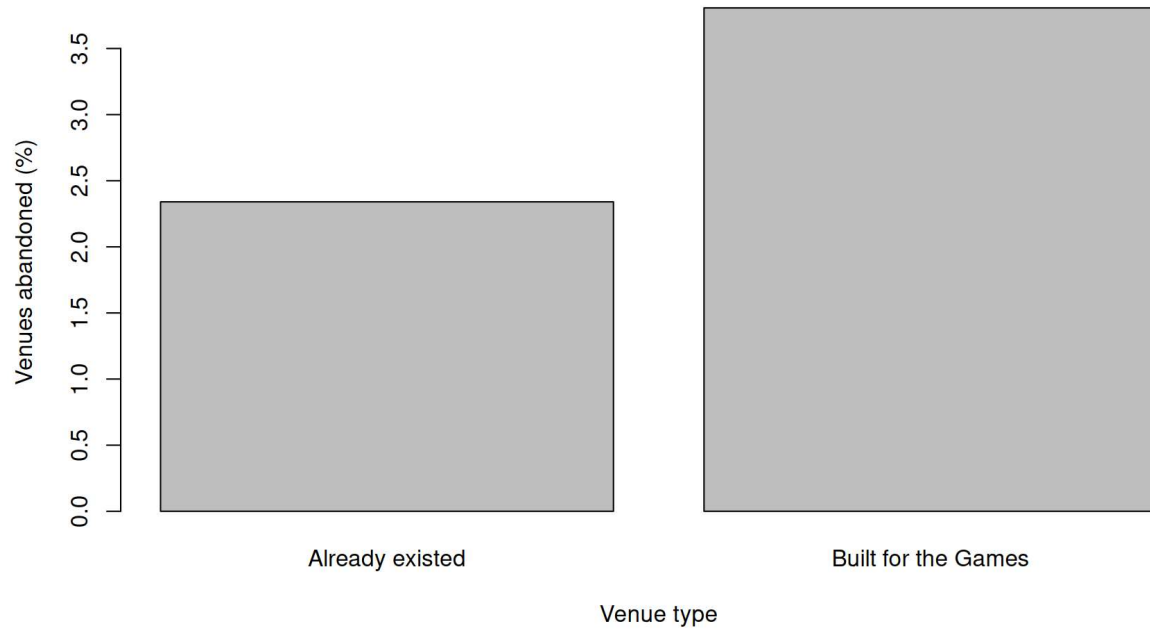


Figure 3. Share of venues abandoned, by whether the venue was built for the Games or already existed. Eleven venues carry no build type and are excluded.

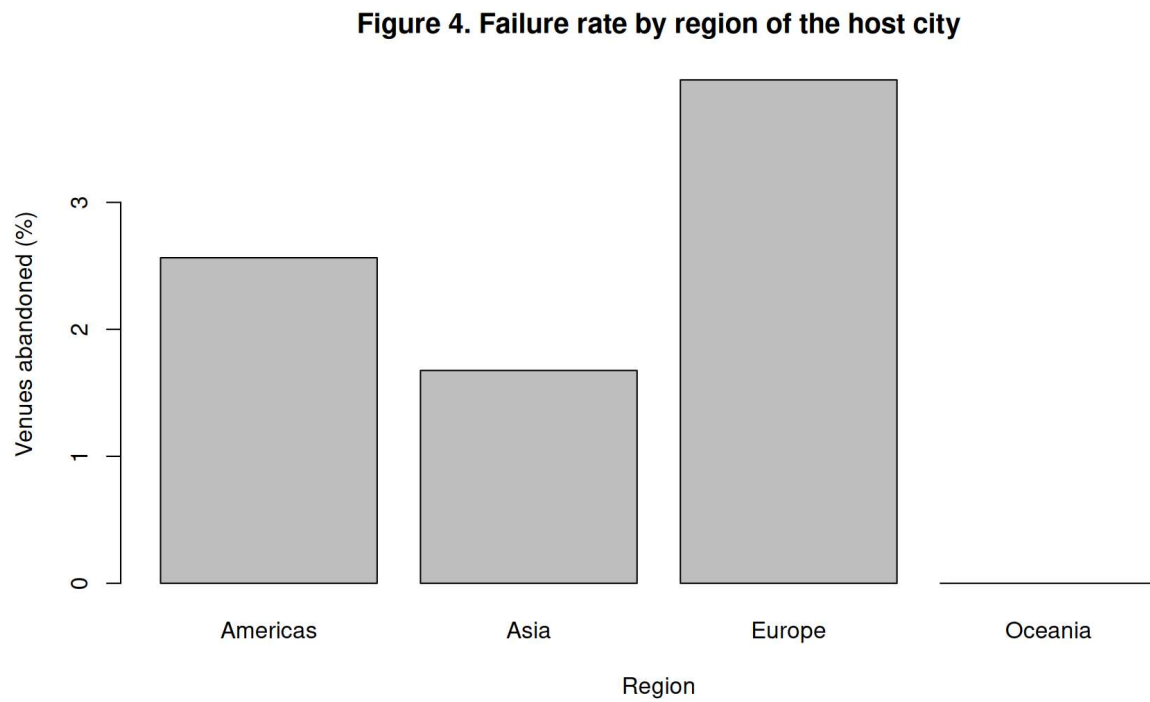


Figure 4. Share of venues abandoned, by region of the host city.

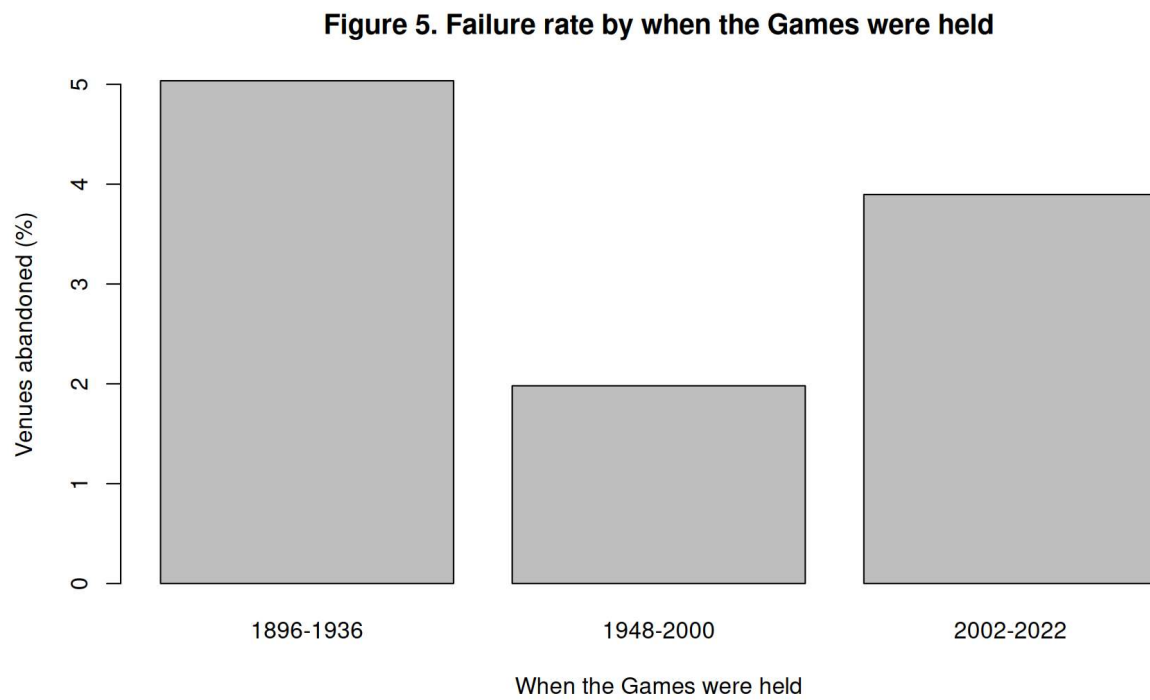


Figure 5. Share of venues abandoned, by the era in which the Games were held.

Table 1. Olympic venue records from the raw pull through to the modeled sample and its train/test split.

Stage	Venues	Failures	Failure rate
Olympic venue records pulled	983	n/a	n/a
Temporary venues, no outcome coded	108	n/a	n/a
Analytical sample	875	26	3.0%
Training set	567	17	3.0%
Test set	308	9	2.9%

Table 2. Share of each predictor missing before imputation, for the analytical sample and for the training rows the imputation medians were taken from.

Variable	Missing, all 875	Missing, 567 training
Log capacity	84.5%	83.8%
Venue age	80.8%	80.6%
Years since construction	80.8%	80.6%
Log distance from center	77.4%	77.4%
Log GDP	38.2%	39.0%
GDP percentile rank	38.2%	39.0%
Log city population	13.5%	12.5%
City population percentile rank	13.5%	12.5%

Variable	Missing, all 875	Missing, 567 training
New-build status	1.3%	1.1%
Years since event	0.0%	0.0%
Region	0.0%	0.0%

Models

This section presents the project's approach to dimensionality reduction, feature selection, and model development. We begin with an unsupervised exploration of the feature space, followed by two untuned baseline models and three tuned classifiers. The section closes with a comparison of model performance and an assessment of whether any model is ready for use. The Elastic Net / Penalized Logistic Regression and Thresholds as a Range sections that follow build on this same modeling framework, using the predictor set and cross-validation design described here.

Pre-processing and Dimensionality Reduction

Feature selection proceeded in two passes rather than one. The first pass, described in the Data section, used a per-predictor missingness check (Fisher's exact test against the outcome) to decide what could safely be imputed; venue age failed that check and was dropped. The second pass was the unsupervised exploration below, run to see whether the feature space could be safely compressed before any classifier was fit. Both passes were run on training rows only, using the same principle applied throughout preprocessing: nothing derived from the test set should shape how the model is built.

Unsupervised Exploration: PCA and K-Means

Fitted model

Before fitting any classifier we ran principal component analysis and k-means clustering on the Olympic training rows, to find out whether the feature space could be compressed first. We fitted both methods on the 567 training rows only, for the same reason the imputation medians were fitted there, and gave neither method the outcome. We brought the outcome in afterwards, only to check whether the structure the two methods found had anything to do with what we are predicting.

Seven imputed numeric columns went in: log capacity, log GDP, log city population, log distance from the city center, venue age, years since the event, and years since construction. Those seven are wider than the five predictors the classifiers ended up using, which is the point of running the step where we ran it. Our question was whether a wider set needed compressing, and we asked it before venue age was dropped and before region was added.

What PCA found

The components split the variance almost evenly instead of concentrating it. The first explains 26.1% and the second 19.9%, and five of the seven are needed to reach 87.5% (Figure 6).

Trading seven columns for five components is not a simplification worth making, and the trade would cost us the ability to say what the model is reacting to. Our stakeholder needs that ability more than it needs a smaller model, because a bid committee can question a seating capacity but has no way to question a component that mixes capacity, distance and age together.

The loadings say the same thing. Venue age (0.66) and years since construction (0.58) dominate the first component, which makes it closer to an age axis than to anything about the venue itself. The second sets years since the event (0.55) against distance (-0.48) and capacity (-0.36). The third is mostly GDP (0.77) against city population (-0.52). No component among the three corresponds to a quantity a planning office could name or act on.

What the clustering found

Clustering pointed the same way. Within-cluster sum of squares falls smoothly from 3,962 at one cluster to 1,437 at eight, with no elbow anywhere along it, so the data doesn't pick a number of groups for us. We took three for inspection rather than because three was indicated. The groups come out at 25, 236 and 306 training rows, with failure rates of 0.0%, 3.4% and 2.9% against a training baseline of 3.0% (Table 3). Which cluster a row falls into tells us close to nothing about whether that venue goes unused. Our smallest group contains no failures, but at a 3% baseline we would expect fewer than one failure in a group of 25 rows, so the zero is not a result.

Time is what the clusters do separate. Cluster 1 holds old venues that already existed long before their Games, averaging 47 years of age at the event and 88.5 years since construction, and sitting farther from the city center than the other two groups. Clusters 2 and 3 both hold venues that were new or near-new at their Games, and the two groups divide on how long ago the Games were, 24 years against 71. The dominant structure in this feature space is therefore when a venue was built and used, rather than what kind of venue it is. Our hypothesis is about venue design, and design is not what these seven columns mostly vary on.

One substantive hint runs with the hypothesis rather than against it. Our group of old, pre-existing venues is the group with no failures in it. We are not going to lean on the zero, because

25 rows at a 3% baseline cannot carry the weight, but the direction is consistent with new-build status mattering and gives us a reason to keep that predictor in the set.

What it changed

The step produced a by-product that mattered more than its main result. Checking the three time features against each other showed they are additive by construction, in that years since construction equals years since the event plus the venue's age at the event. Because we impute the three columns independently against training medians, the relationship survives on only 110 of the 567 training rows. Finding that is what sent us to look at venue age's missingness directly, and venue age was dropped for the reason given in the Cleaning section.

Nothing else was carried forward. No components replaced the original columns and cluster membership was not added as a feature, because neither buys enough to pay for what it costs in interpretability. Our reasoning for not returning to PCA later, as a model candidate rather than as a simplification, is in Reconsidering PCA as a Model Candidate.

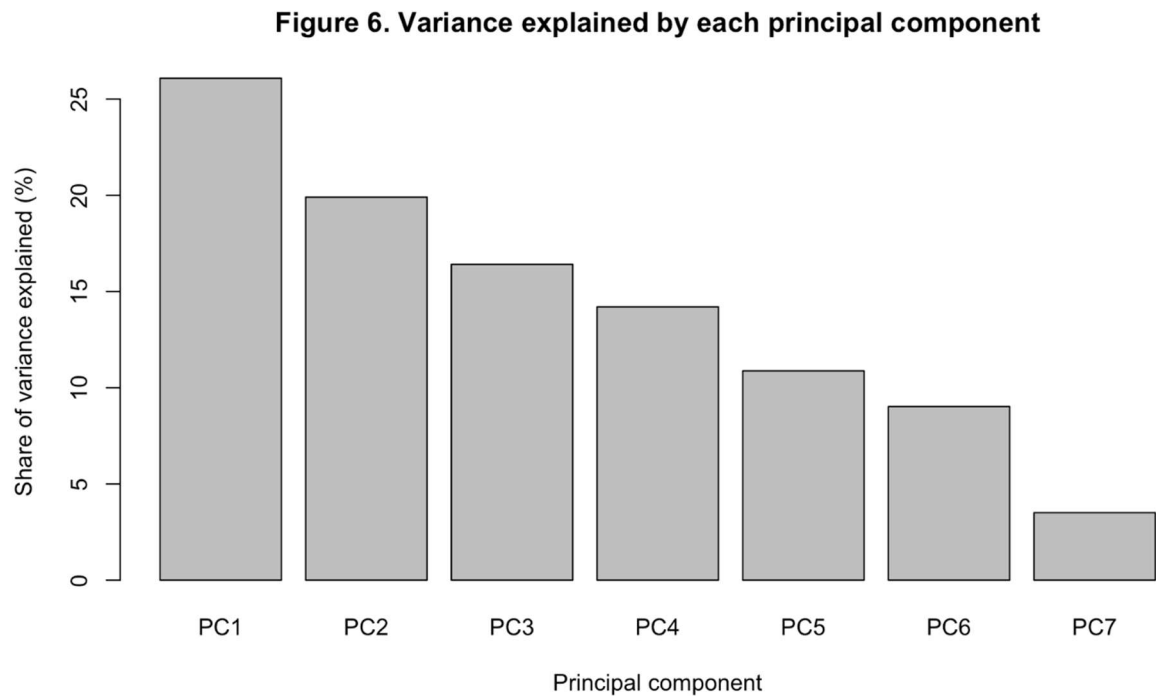


Figure 6. Share of variance explained by each of the seven principal components, Olympic training set ($n = 567$ rows). The bars fall gradually rather than dropping away after the first two, which is what makes the components a poor replacement for the original columns.

Table 3. Average profile of each k-means cluster on the seven input features, Olympic training set ($n = 567$ rows), with the share of each cluster that became a white elephant. Capacity, GDP, city population and distance are shown on the log scale they were clustered on. The three age columns are in years. Cluster numbers are assigned arbitrarily by the algorithm and carry no order.

Cluster	Rows	Failure rate	Log capacity	Log GDP	Log city pop.	Log distance	Venue age	Years since event	Years since construction
1	25	0.0%	10.03	9.12	7.46	2.84	47.1	41.4	88.5
2	236	3.4%	9.59	9.67	7.88	1.96	6.6	24.0	38.2
3	306	2.9%	9.60	8.49	7.87	1.93	6.4	71.3	43.2

Baseline Models: Logistic Regression and LDA

Fitted Model

Two untuned baselines were fit on the Olympic training data: logistic regression, chosen because the outcome is binary and its coefficients are easy for a non-technical stakeholder to read, and linear discriminant analysis (LDA), added as second baseline because it makes different assumptions and gives a useful point of comparison. Both were fit on the original four-predictor set (log capacity, GDP percentile rank, host-city population percentile rank, and new-build status; region was added later, in Sprint 6) and scored with 5-fold cross-validation repeated 10 times, using folds assigned at random rather than grouped by Games edition, since the leakage this created was not identified until feedback on this report.

Logistic regression averaged an AUC of 0.586 across the 50 folds (range 0.292 to 0.861); LDA averaged 0.597 (range 0.422 to 0.828). Both land only slightly better than guessing. At each model's Youden-optimal cutoff, logistic regression reached 97.1% sensitivity and 24.5% specificity (balanced accuracy 60.8%), and LDA reached 98.2% and 24.0% (balanced accuracy

61.1%); both catch nearly every real failure only by flagging roughly three-quarters of all venues at-risk, a direct consequence of how weak the signal is at this stage.

Assumptions

Logistic regression and LDA both assume no two predictors carry the same information twice; the Sprint 3 EDA's correlation check on the World Cup numeric predictors found no pair above 0.7, and with only four predictors in the baseline, redundancy was not treated as the likely bottleneck. LDA additionally assumes equal covariance across the two outcome groups and roughly normal predictors; neither was formally tested, since with only 17 training failures there is not enough data on the failure side to test either meaningfully. If covariance looks unequal once the predictor set is finalized, QDA is the natural comparison to run.

Overfitting

Cross-validation was the main control at this stage, along with keeping the predictor count low relative to the 17 available failures (roughly 4 failures per predictor, below the usual floor of 10). Testing and removing venue age also functioned as an overfitting control, since it stopped the model from exploiting a pattern that reflected Wikidata's coverage rather than the venues themselves; re-running with venue age swapped out for host-city population barely moved the AUC (0.588/0.592 with venue age versus 0.586/0.597 without), confirming venue age was never contributing real signal. These baselines were later found to be validated under folds that let a Games edition split across training and validation (see Project-Specific Requests); the AUCs above are the original, ungrouped-CV numbers and are treated as an upper bound rather than a clean estimate.

Elastic Net / Penalized Logistic Regression

Fitted Model

All 330 combinations of alpha and lambda tested were checked, not just the one that came out on top, to understand why it landed where it did. None beat random guessing. 131 combinations were regularized enough to zero out every predictor, and all of those scored exactly 0.500 (chance level). The other 199 kept at least one real predictor, and every one of those scored below chance, as low as 0.411 (see Project-Specific Requests for a direct check confirming this below-chance result reflects real instability from the small failure count rather than a coding error). Even the weakest regularization tested, which kept all seven predictor terms, only reached an AUC of about 0.44.

The model that zeroed everything out was not tied with a group of equally useless options, it was the best of everything tested, because every version that kept real predictors did worse than a coin flip. This indicates something about the predictor set itself: these five predictors do not just fail to help, using them actively makes predictions worse. At the cutoff normally used to turn probabilities into yes/no predictions, the model breaks down completely and cannot produce a usable cutoff (see Thresholds as a Range for a full accounting across all three tuned models), so predictions land close to a coin flip, correctly catching about half of true failures while correctly clearing about half of safe venues.

Compared against the Sprint 5 baseline logistic regression and LDA models (AUC 0.586 and 0.597 respectively, both computed under ungrouped cross-validation), the elastic net's grouped-CV AUC of 0.500 represents a real decline once Games-edition leakage was corrected, not a failure of tuning. The original baseline numbers are read as inflated by that leakage rather than reflecting genuine predictive signal.

Assumptions

Elastic net assumes the predictors relate to the outcome in a straight-line (linear, on the logit scale) way, and that observations are independent of one another once those predictors are accounted for. Grouping the cross-validation folds by Games edition addresses the independence assumption for how the model is tested; the original train/test split may still carry a similar issue, addressed under Discussion and Next Steps. World Cup venues were evaluated separately with a simple, non-cross-validated model, the same approach used for the baseline models, because only three to four total failures exist on that side, not enough to cross-validate.

Overfitting

Overfitting was controlled through the grouped, repeated cross-validation described in Methods, plus the shrinkage built into elastic net itself. Here, that shrinkage pushed the model all the way down to using no predictors at all. This is read as the model correctly refusing to rely on predictors that do not hold up under grouped validation, rather than a sign the overfitting controls failed.

Random Forest

Fitted Model

We fitted a random forest on the Olympic training rows, using the same five predictors as the other two tuned models: log capacity, GDP percentile rank, host-city population percentile rank, new-build status, and region. Keeping the predictor set identical across all three is what makes them comparable to one another. It also means the forest carries one predictor the baseline models did not have, since region was added at the tuning stage.

The forest was fitted on 561 of the 567 training venues. Six venues have no recorded build type, and the forest drops any row with a missing predictor value, which removes those six. None of them is a failure, so all 17 training failures survive, spread across 53 Games editions.

The forest has one setting worth tuning. That setting fixes how many columns the forest can draw on at each split. Region records which of four continents a venue sits on, and it enters the model as three yes/no columns, so the forest sees seven columns rather than five predictors. We compared a setting of 3, the rounded default for seven columns, against 7, which makes every column available at every split. Both results appear in Table 4, scored under grouped folds and under ordinary ones on the identical rows.

Under grouped folds the forest scored 0.403 and 0.416, below chance at either setting. The difference between the two settings is far smaller than the fold-to-fold variation, which runs between 0.12 and 0.13, so the tuning made no difference we can act on. The forest does not separate failed venues from surviving ones.

Variable importance is mixed rather than uniformly flat, and it does not point where our hypothesis expected (Figure 7). Each column is scored by how much worse the forest gets when that column is scrambled. GDP percentile rank and the Europe column score highest, at 2.9 and 3.5. New-build status scores lowest at -4.7, meaning the forest did slightly better with that column scrambled than with it intact. And our hypothesis names new-build status as one of its two strongest predictors. These figures come from the better-scoring of the two forests, at 0.416, so all of them sit inside a model that cannot separate the two outcomes. We read the ordering rather than the sizes, and we do not read much into either.

Assumptions

A forest carries fewer assumptions than the baseline models, which is why it was chosen. There is no assumption about the shape of the relationship between a predictor and the outcome, no need to scale anything, and the imputed columns go in as they are. Nothing in the model requires the predictors to be uncorrelated.

The assumption that does bind is independence, and it binds on how the model is tested rather than on the model itself. Venues from the same Olympics share one GDP figure and one host-city population, so folds built at random put the same values on both sides. Cross-Validation Leakage covers that in full, and the same clustering is why the two fold designs in Table 4 give such different numbers.

At the default cutoff the forest labels every venue as not failed and catches none of the 17 failures. At a 3% failure rate that is the expected behavior rather than a defect, and the baseline models did the same thing.

Overfitting

Grouped repeated cross-validation is the control, and it does two jobs. Grouping keeps every venue from one Olympics inside a single fold, and repeating the whole procedure ten times gives a range rather than one number that a single unlucky split could have produced.

We report that range for the same reason the baseline report did. With 17 failures spread across five folds, any one fold is scoring three or four real cases, so a single fold's score carries almost no information on its own.

The forest was not scored on the held-out test set. Discussion and Next Steps explains why that was left out rather than reported.

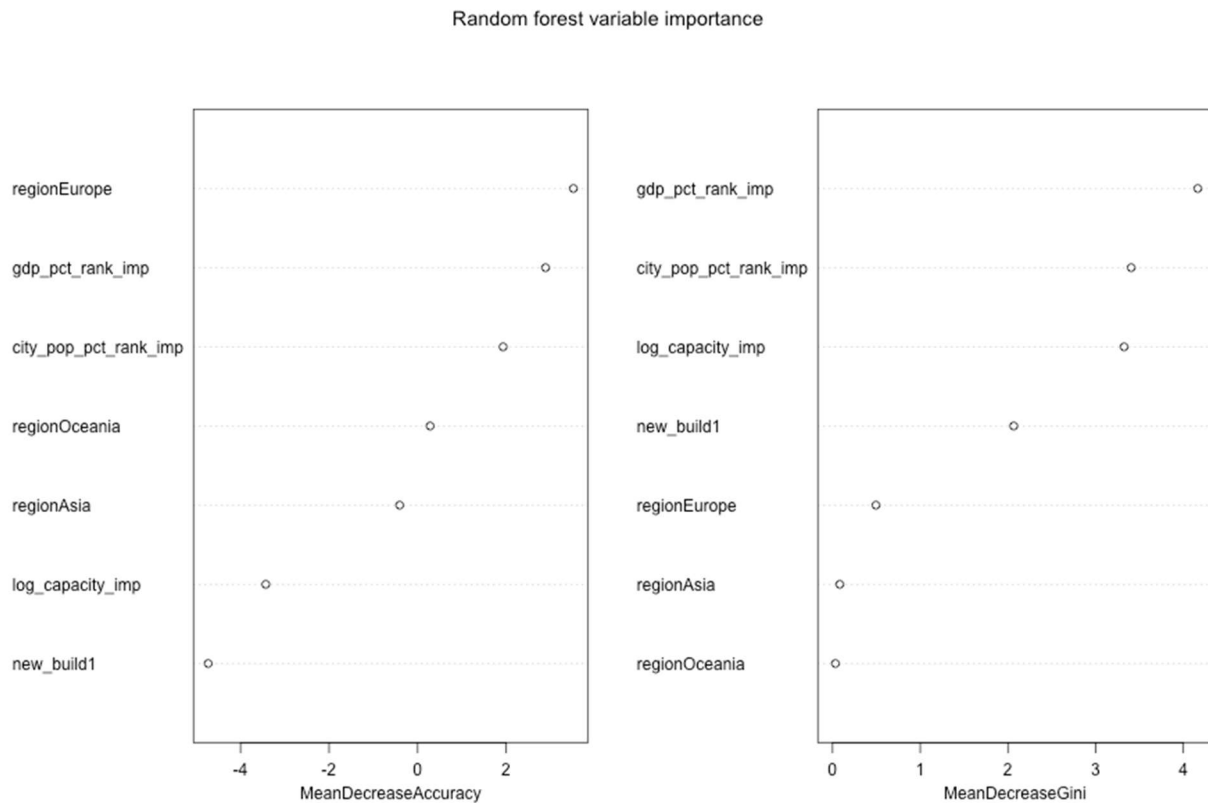


Figure 7. Random forest variable importance, from the better-scoring of the two settings tested.

The ordering is readable; the sizes sit inside a model scoring below chance and should not be read as effect sizes.

Gradient Boosting

Fitted Model

The best gradient boosting model used 250 trees, a tree depth of 1, a learning rate of 0.10, and a minimum node size of 10. It was trained on the same five predictors used by the other Sprint 6 models: log capacity, GDP percentile rank, host-city population percentile rank, new-build

status, and region. The model was evaluated using grouped 5-fold cross-validation repeated 10 times, which kept all venues from the same Games edition in the same fold to prevent the edition-level leakage identified earlier in the project.

The model produced a mean cross-validated AUC of 0.492, which is approximately chance-level performance. Across the 50 resamples, AUC ranged from 0.127 to 0.865 with a standard deviation of 0.174. This wide range shows that the model's performance was unstable across folds, which is likely influenced by the small number of white-elephant cases in the training data. With only 17 failures in total, each validation fold contains only about three to four failures. Variable importance ranked GDP percentile as the most influential predictor, followed by log capacity and host-city population percentile. New-built status and the Europe region indicator contributed much less, while the Asia and Oceania region indicators had no measured importance. These results should be interpreted carefully because the model's overall AUC was below 0.50. A predictor can still appear relatively important within a model even when that model does not provide useful predictive performance.

Assumptions

Gradient boosting does not require the predictors to have a linear relationship with the outcome, follow a normal distribution, or be scaled before modeling. The more important assumption for this project is that the observations used for validation are independent from those used for training. Venues from the same Olympic edition share the same GDP and host-city population values, so grouping the cross-validation folds by Games edition was necessary to prevent those shared values from appearing on both sides of a fold. Gradient boosting also builds trees sequentially, with each tree correcting errors from earlier trees, which can make the model

sensitive to a small number of influential observations. That is especially important here because only 17 white elephant cases are available for training.

Overfitting

Overfitting was controlled through grouped repeated cross-validation and by limiting the complexity of the tuning grid. We tested tree depths of only 1 through 3, held the minimum node size at 10, and tested learning rates from 0.01 to 0.10. The selected model used the shallowest tree depth tested, a depth of 1, which suggests that allowing more complex trees did not improve cross-validated performance. Overall, the GBM result was consistent with the elastic net and random forest results. After controlling for Games-edition leakage, the current predictor set did not provide enough generalizable information to reliably distinguish white-elephant venues from venues that remained in use.

Model Tuning and Validation: Comparing All Five Models

Every model above was scored the same way, repeated 5-fold cross-validation, so the numbers are directly comparable. The table below lines up all five models scored on the Olympic training data; the elastic net's fitted-model detail is in the Elastic Net / Penalized Logistic Regression section, and a threshold-level comparison of the three Sprint 6 models is in Thresholds as a Range.

Model	AUC (grouped, repeated CV)	vs. chance (0.500)	Notes
Logistic regression (S5 baseline)	0.586	above	Ungrouped CV; likely inflated by Games-edition leakage
LDA (S5 baseline)	0.597	above	Ungrouped CV; likely inflated by Games-edition leakage
Elastic net / penalized logistic (S6)	0.500	at chance	Best of the three S6 models, only by discarding every predictor
Random forest, mtry = 3 (S6)	0.403	below	Ordinary (ungrouped) folds gave 0.635 on the same rows
Random forest, mtry = 7 (S6)	0.416	below	Ordinary (ungrouped) folds gave 0.638 on the same rows
Gradient boosting (S6)	0.492	below	Best tuning: 250 trees, depth 1, learning rate 0.10

Table 4. Cross-validated AUC for all five models fit in this project. The two Sprint 5 baselines were scored under cross-validation folds later found to let Games-edition data leak between training and validation; all three Sprint 6 models were scored under folds grouped to prevent that leakage, and the random forest was also scored under both fold designs on the identical rows to measure the leakage directly (0.22 to 0.23 of AUC).

Fixing the Games-edition leakage changed the results substantially. Once folds were grouped correctly, none of the three tuned models beat the Sprint 5 baseline on raw AUC, which is read as evidence that the baseline numbers were inflated by leakage rather than as a sign the tuning failed. Across the three Sprint 6 models, elastic net's 0.500 is the best score achieved, and it

comes from a configuration that discarded every predictor; every configuration in every one of the three tuned models that kept a real predictor scored at or below chance.

Final/Best Model

No model produced in this project is being called a best or final model in the sense of being ready to deploy. None of the three tuned models beats chance under the corrected, leakage-free cross-validation, and the two untuned baselines that do score above chance were validated under folds now known to leak. Calling the elastic net's 0.500 AUC configuration a best model would be accurate only in the narrow sense that it was the least bad option tested; it predicts using no information about the venue at all, which is not a usable classifier for a host-city bid committee.

The honest final answer, given the current data, is that none of the predictors we can currently test, log capacity, GDP percentile rank, host-city population percentile rank, new-build status, and region, show a useful, generalizable link to whether an Olympic venue becomes a white elephant, once Games-edition leakage is corrected. That is a statement about the current predictor set and the small number of training failures (17), not a verdict on the underlying research question: design purpose, the predictor the hypothesis leans on most heavily, remains unmeasured on the Olympic side and is the leading candidate for closing that gap. In lieu of a deployable model, the required post hoc exploration for this project is in Thresholds as a Range, which shows that no probability cutoff, across any of the three tuned models, produces an operating point a planning office could act on; the closest any model comes to a usable balance is gradient boosting at a threshold of 0.015, with sensitivity and specificity both still below half.

What we would do next, in priority order, is laid out in full in Discussion and Next Steps: build design purpose for Olympic venues from the `use_at_games` column already in the data, re-examine whether region belongs in the predictor set now that it was added under folds that have since been replaced, and regroup the train/test split itself by Games edition before scoring anything on the held-out test set, since that split currently has the same clustering problem the cross-validation folds did.

Thresholds as a Range

Reporting a single classification threshold would understate how weak each model's discrimination actually is, so thresholds were evaluated across a range rather than at one arbitrarily chosen cutoff (Figure 8). None of the three tuned models produced a threshold that both caught most real failures and kept the flagged rate at a usable level. The penalized logistic regression's out-of-fold predictions cluster within a narrow band (0.024 to 0.044), so it has almost no usable threshold range at all, moving directly from flagging every venue as at-risk to flagging none within a span of about two percentage points of probability. The random forest never exceeded 11.2% sensitivity at any threshold tested, meaning even its most permissive cutoff missed about seven of every eight real failures. The gradient boosting model could catch most failures, 85.9% sensitivity at its lowest threshold, but only by flagging 89.1% of all venues in the dataset as at-risk, which is not an actionable cutoff. The closest any model came to a balanced tradeoff was the gradient boosting model at a threshold of 0.015, with sensitivity of 48.8% and specificity of 47.6%, both still below half. Taken together, this reinforces the pattern in Model Tuning and Validation above: the weakness in these models is not a matter of picking the wrong cutoff, no cutoff exists that makes any of them usable for flagging individual venues as at-risk with the current predictor set.

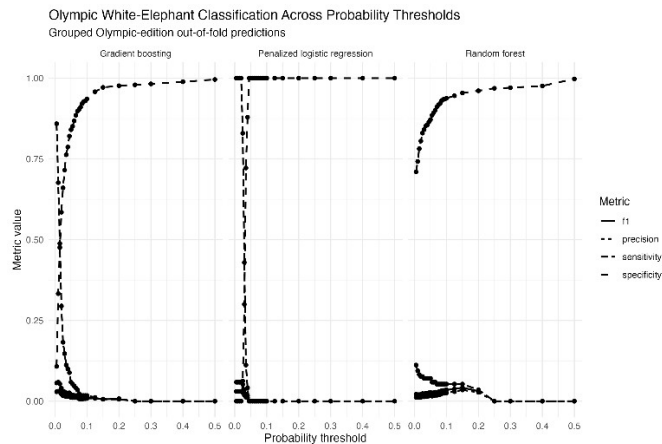


Figure 8. Sensitivity, specificity, precision, and F1 across a grid of probability thresholds from 0.005 to 0.5, for each of the three tuned models, using pooled out-of-fold predictions from the grouped Olympic-edition cross-validation.

No single threshold achieves strong sensitivity and specificity simultaneously for any model

Project Specific Requests

Reconsidering PCA as a Model Candidate

Feedback on the M04 report was otherwise positive but raised one nuanced point. Our PCA results argued against PCA as an interpretability-preserving simplification. But explained variance alone cannot establish whether PCA would actually help or hurt prediction. If dimensionality became a problem during supervised modeling, the feedback said, PCA should be reconsidered as an actual model candidate and compared via cross-validation. Dimensionality never became that kind of obstacle. Every tunable model we fit settled on four to five predictors: log capacity, GDP percentile rank, city population percentile rank, new-build status, and region. We kept that set small on purpose, since we had only 17 training failures to learn from. If anything, our constraint ran the opposite direction. We held back predictor candidates rather than

accumulating too many. We did not revisit PCA as a model candidate because the situation this feedback anticipated never arose.

Cross-Validation Leakage

Feedback on the M05 report flagged a problem with our cross-validation folds. Because they were assigned randomly, venues from the same Games edition could split across training and validation. GDP and city population are identical for every venue at a given Games, so this meant models were partly validated on values they had already seen in training. We fixed this by building the folds ourselves. A custom function keeps every row from the same Games edition in a single fold. It also balances failure counts across folds as evenly as the grouping allows. This grouped, repeated design, 5 folds times 10 repeats, was applied to all three models tuned in Sprint 6. It materially changed our results. Once folds were grouped correctly, our elastic net's cross-validated AUC dropped from baseline range (0.586 to 0.597) to 0.500 at best. We read that as evidence the original baseline numbers were inflated by the leakage identified in that feedback.

Verifying a Below-Chance Result Was Not a Reversed Ranking

Feedback on the S6 report noted that our elastic net's 0.41 AUC was not a mistake, but was worth verifying. Inverting the predicted probabilities would give 0.59, so the feedback asked us to rule out a systematically reversed ranking. We checked three things directly. First, we confirmed our outcome variable's factor levels are ordered correctly. The target class genuinely represents a white-elephant venue, not the reverse. Second, our ROC calculations already fix direction explicitly rather than letting it be auto-detected from the data. That's the same class of bug we caught and fixed in our Sprint 5 baseline models, so this isn't a repeat of it. Third, and

most directly, we isolated the specific model configuration that produced the 0.411 AUC. We then compared its actual out-of-fold predictions between true failures and true survivors. True failures averaged a 2.8% predicted risk, versus 3.5% for true survivors. That confirms the below-chance ranking is real in the predictions themselves, not a byproduct of how AUC was computed.

We don't read this as a systematically reversed, generalizable relationship worth flipping and reporting as 0.59. The gap between 2.8% and 3.5% is small, and both numbers sit close to our roughly 3% base rate. This configuration also uses comparatively light regularization on a training set with only 17 total failures. Each cross-validation fold sees just three or four real cases. We read this as instability from too little data, not a real inverse signal. A lightly regularized model can fit idiosyncratic patterns in one fold's small set of failures. That can rank a few of them as lower-risk without the pattern generalizing. This is consistent with what we already concluded elsewhere in the report. The configuration selected by tuning zeroed out every predictor entirely. We read that as the shrinkage correctly refusing to rely on a signal that isn't there.

Region as a Predictor: Confirming the Effect Isn't Measurement Noise

Feedback on our Sprint 6 individual reflections asked a clarifying question. Was the 0.05 to 0.07 of AUC we reported for region an absolute difference between two fitted models, or a difference measured across folds? The feedback also gave us a decision rule. If a confidence interval on the paired mean delta excludes zero, the answer is clear in whichever direction it falls. If the interval overlaps zero, there is no evidence either way, and we decide on a substantive argument instead. The figure is measured across folds. We re-ran the random forest on four predictors and on five, and both fits used the same single set of grouped folds. That means each of the 50 resamples,

five folds repeated ten times, pairs a with-region model against a without-region model on the same rows. Every figure in our Sprint 6 table is a mean AUC across those 50 resamples.

Subtracting one row from the other gives the paired mean delta at each mtry. Adding region costs 0.050 of AUC at mtry 2, and 0.066 at mtry 4.

We haven't put a confidence interval on that mean. The reason is the same clustering problem we ran into with the folds and with the train/test split. The 50 resamples are ten reshuffles of the same 561 training rows. Each training set is four-fifths of those rows, so any two training sets share most of their rows, and their scores move together. An interval that treats those scores as independent comes out narrower than the evidence supports. Running more repeats wouldn't help either, since the problem is the overlap between training sets, not the number of resamples.

So we're in the third case of that decision rule. We're keeping region for the reason we gave last sprint. Having all three models on one predictor set is worth more than a fifth predictor whose effect we can't separate from noise. Region also has no missing values at all. Capacity, by comparison, is imputed for 84% of the training rows, and venue age came out of the predictor set entirely.

Discussion & Next Steps

Conclusions

We asked whether information a host city already has at the planning stage can predict which venues end up abandoned. On the evidence we assembled, it can't. No model we built separates failed from surviving venues once the cross-validation folds respect the Games-edition grouping. The best score any tuned model reached was 0.500, and that came from a configuration that

discarded every predictor (Table 4). A model that gives every venue the same score can't rank anything, so that number is just arithmetic. We don't have a best model here in the sense of one we could hand to a planning office.

Our hypothesis named three features: newly built status, single-purpose design, and distance from the city center. Only the first one is in the data. The exploration shows it pointing the way the hypothesis expected, with venues built for the Games failing at 3.8% against 2.3% for venues that already existed (Figure 3). No model picked that difference up as usable signal, and the gap rests on 15 failures against 11. Our prediction said newly built status and design purpose would come out as the strongest predictors, and that the model would beat a naive baseline. Neither happened. New-build status scored lowest of all the predictors in the random forest, at -4.7. And no model beat a rule that calls every venue safe, which at a 3.0% failure rate is right 97% of the time.

The way the prediction failed tells us something about where the problem sits. Three models built on different principles agree on the same five predictors: a random forest at 0.403 to 0.416, a gradient boosting model at 0.492, and a penalized regression that got its best available score by throwing all five predictors away. When three different model classes fail the same way on one predictor set, that points at the predictors and not at the models.

The result in context

A negative result needs context to be worth anything. Ours doesn't contradict the earlier work; it sits at a different stage of it. Alm et al. (2016) found that ownership, capacity, and corruption mattered for post-event venue use, but they measured venues after the fact, and every venue in that study was newly built, so their design can't separate new from existing venues the way our

question needs. Müller (2015) describes the mega-event syndrome as a process failure in how host cities plan. That's an account of why venues go wrong, not a venue-level predictor of which ones will. Flyvbjerg et al. (2016, 2024) establish that the budget record is poor, but cost overrun and post-event disuse are different outcomes. None of that work claims venue-level prediction from pre-construction information is possible, and our result is the first evidence we're aware of about whether it is.

What we contribute is a negative result with a diagnosis attached to it. The obstacle isn't that the relationship is absent. It's that the planning-stage record for Olympic venues is too thin to test it. Seating capacity is unrecorded for 84.5% of the sample (Table 2), and design purpose isn't recorded for Olympic venues at all. Those two features mattered most to our hypothesis, and they're the ones we could measure least well. For anyone attempting this question next, the practical conclusion is about data collection: these columns need to be recorded before the question can be modeled at all.

One methodological result applies beyond this project. We scored the same random forest on the same rows twice, once under ordinary folds and once under folds grouped by Games edition, and the difference is 0.22 to 0.23 of AUC (Table 4). Venues from one Games share a single GDP figure and a single host-city population, so ungrouped folds let a model see values in validation that it had already seen in training. Any study modeling venues, matches, or projects that cluster inside an event should expect an inflation of about that size if it resamples rows independently.

Recommendations

For our stakeholder, the recommendation is not to use any model in this report to screen a proposed venue. None of them is good enough for that. The threshold analysis also shows the

problem isn't a badly chosen cutoff, because no cutoff makes any of the three tuned models usable.

The recommendation we can make is about data, not models. Design purpose and seating capacity are both known to a bid committee at the moment a venue is proposed, and neither one is recorded in any public source we could find covering Olympic venues historically. If a body recorded them at bid submission, going forward, this question would be answerable within a decade of hosting cycles. That's a cheaper fix than any modeling improvement we could suggest.

If we continued the work ourselves, in priority order:

1. Build design purpose for the Olympic venues. The Olympic table already has a column recording the sports each venue hosted, filled in for all 875 rows, and counting the sports listed separates single-purpose from multi-purpose venues without needing a new data source. Two caveats apply. The column isn't built yet. It also records what a venue was used for during the Games, not what it was designed for beforehand, so it measures something close to the World Cup column but not the same thing. The difference would have to be stated wherever the two arms are compared.
2. Re-examine whether region belongs in the predictor set. We added it under the fold design we've since replaced, and under grouped folds it costs the forest 0.050 to 0.066 of AUC instead of adding anything. The gap is smaller than the fold-to-fold spread, so this doesn't establish that the effect is real. The evidence only says that nothing supports having added it.
3. Regroup the train/test split by Games edition before scoring anything on the held-out test set. The split currently groups by venue, so venues from one Olympics sit on both sides

of it. That's the same clustering problem the folds had. Nothing we report rests on the test set, because no script evaluates on it, so this is a fix to make before the first test-set score, not a correction to anything we've published.

Caveats and limitations

The largest limitation is that our most-used continuous predictor mostly isn't a measurement. Seating capacity is absent for 84.5% of the analytical sample, and imputation fills every one of those rows with the same training median, so the column the models see is a real value for fewer than one venue in six and a constant for the rest (Figure 2). Its standard deviation falls from 1.14 among observed values to 0.45 across the column as a whole. Any importance the models assign to capacity is therefore partly an artifact of which venues happen to have a Wikidata record, and we wouldn't defend a capacity effect on this data. The same concern applies with less force to distance from the city center at 77.4% missing and to host-country income at 38.2%. With venue age it applied with enough force that we dropped the variable outright, once a test showed every one of the 17 training failures had it missing.

Four further concerns we couldn't resolve. The class is small in absolute terms: 26 failures, of which 17 are in the training rows, so any single fold is scoring three or four real cases, and the fold-to-fold spread of 0.12 to 0.13 is what that arithmetic produces, not a property of the models. Our sample is geographically lopsided, with 454 of 875 venues in Europe and 47 in Oceania (Figure 4), so a regional effect is mostly a European effect, and the zero for Oceania rests on too few venues to mean anything. The target is right censored: it records each venue's status as of data collection, and a venue in active use today can still fail in ten years. The bias runs toward calling recent venues successes, and a model fitted on this data can't correct for it. And the

Tokyo Olympic Village is two distinct buildings sharing a name and a host city, which the current split treats as one venue.

Two sources of bias sit in the target itself and not in the predictors. We hand-coded the outcome from status and reason text, and the boundary between a venue abandoned before demolition and one lost to urban redevelopment is a judgment call, not a fact in the source. 69 venues sit on the redevelopment side against the 26 we count as failures, so moving that boundary would change the target substantially, and an earlier definition that counted any venue not in use gave 106 failures instead of 26 (Figure 1). We don't think our coding is wrong, but a reader should know the outcome is a definition we chose and defended, not one the data supplied. The second concern is simpler. We have no unbiased estimate of out-of-sample performance for any model in this report, because the held-out test set has deliberately never been scored. Every figure we give comes from cross-validation inside the training rows.

Code Availability Section

All code, data processing scripts, and modeling scripts for this project are available at the team's GitHub repository:

<https://github.com/Smoothspinner/sports-urbanism-capstone>

The repository README documents the full run order, from raw data cleaning through final model tuning, and lists every custom function used along with what it does and where. To reproduce this analysis: open RStudio, set the working directory to the repository root (all scripts use paths relative to that root), and install the required packages: tidyverse, countrycode, geosphere, stringi, httr, jsonlite, readxl, WDI, caret, pROC, glmnet, and gbm (MASS installs automatically with caret). All modeling scripts fix a random seed of 123 for reproducibility.

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Appendix

Data dictionary

The data dictionary and codebook is maintained in the team repository and is current as of this submission:

https://github.com/Smoothspinner/sports-urbanism-capstone/blob/main/reports/cleaning_preprocessing/data_dictionary.md

It documents every column in both modeling tables: `olympic_model_split.csv` at 875 rows and 78 columns, and `worldcup_model_split.csv` at 265 rows and 73 columns. The dictionary covers identifiers and source fields, country and city context, engineered features, the rank and standardized versions of each, the split and missing-value columns, and the columns added by the unsupervised step. It replaces the EDA-stage dictionary, which describes the joined tables as they stood before feature engineering, splitting and imputation.

Methods tried and set aside

None of this reached the final models, but we did the work, so it's recorded here.

1. **Principal components and k-means clustering**, run before any classifier to see whether the feature space could be compressed. Neither one was carried forward. The full account is in Unsupervised Exploration.
2. **Venue age**, tested as a predictor and then dropped. All 17 of the training failures had a missing venue age, against none of the 110 venues with a known one, which suggested the variable was really measuring whether a Wikidata record existed. We replaced it with

host-city population percentile rank and the models barely moved, so it doesn't look like it was carrying anything.

3. **The corruption perceptions index**, which we planned to use from the proposal onward and dropped in Sprint 4 because the index only starts in 2012, after most of the venues in the sample were built.
4. **A capacity check** on the penalized regression, asking whether dropping capacity in favor of region cost us real signal, or whether the penalty was shrinking capacity toward zero anyway. We decided to keep both, which is why the shared predictor set has five predictors. That check ran under the fold design we've since replaced, which is why region comes up again in Recommendations.
5. **Scoring the models on the held-out test set**, which we deliberately didn't do. The split groups by venue instead of by Games edition, so it carries the same clustering problem the folds did. Regrouping it changes which rows are in training, and that would move every number in this report. It's a fix to make before the first test-set score, not after.